CS3235 WWLLMS · SECTION D · CIE3 LAB ASSESSMENT

# Multilingual Lecture-to-Notes & Quiz Generator

WWLLM Lecture Studio: Project 5 of the AI4Bharat and BharatGen mini-projects





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github.com/Tamishsri/WWLLM-CIE-3

# 1. The problem statement

The CIE3 lab assessment asks each student to build one of five projects on India's sovereign models and datasets from AI4Bharat and BharatGen. The objective is a usable, end-to-end Indian-language AI application built on pretrained models, with a suitable frontend and measurable evaluation. This submission is Project 5.

> ### 5. Multilingual Lecture-to-Notes & Quiz Generator
>
> | | |
> |---|---|
> | Application | Transcribe a recorded lecture, create clear notes and five MCQs, and optionally translate the notes. |
> | Models / data | AI4Bharat IndicConformer or BharatGen speech model; Param1; IndicTrans2 |
> | Concepts | ASR, summarization, quiz generation |
> | Evaluation | WER, factual consistency, MCQ quality |
> | All projects | 30 to 50 representative test cases, a failure analysis, a usable frontend and a GitHub repository |
>
> CS3235 Mini-Project Ideas, October 2026

A recording is slow to revise from and can't be searched. In India it may also mix languages, or be given in a language the student reads less easily. This project turns one recording into material a student can read, test themselves on and return to in their own language.

# 2. What the brief asked for and what was delivered

| The brief asks for | Delivered | Status |
|---|---|---|
| Transcribe a recorded lecture | Silero-VAD and Whisper on uploaded audio or video | Met |
| Clear notes | Structured English notes from Param-1, streamed as they are written | Met |
| Five MCQs | Five questions from Param-1 by default, adjustable from 3 to 10 | Met, quality mixed |
| Optional translation | Notes and quiz in ten Indian languages | Met |
| IndicConformer or a BharatGen speech model | Whisper. The IndicConformer code path is in place but unrun, because the model is gated | Substituted |
| Param1 | BharatGen Param-1 2.9B Instruct | Met |
| IndicTrans2 | NLLB-200. The IndicTrans2 code path is in place but unrun, because the model is gated | Substituted |
| WER | Measured on 40 lectures: average 0.258 | Met |
| Factual consistency | Hand check of the sample notes: 16 of 17 statements correct | One lecture only |
| MCQ quality | Hand check of the sample quiz: 1 of 5 questions fully sound | One lecture only |
| 30 to 50 test cases | 40 lectures with reference transcripts | Met |
| Failure analysis | Section 7, and `outputs/evaluations/failure_analysis.csv` | Met |
| Usable frontend | A custom web studio, plus a Streamlit app | Met |
| GitHub repository | github.com/Tamishsri/WWLLM-CIE-3 | Met |

| Assessment rubric | Where it is covered |
|---|---|
| Problem definition & relevance | Section 1 |
| Model / dataset selection | Section 4 |
| Working end-to-end application | Sections 3 and 5 |
| GenAI / RAG / agentic integration | Param-1 generates the notes and quiz (Section 4). There is no RAG or agent |
| Quantitative testing & error analysis | Sections 6 to 9 |
| Demo, documentation & repository | Sections 5 and 11, and the repository README |

# 3. What was made

WWLLM Lecture Studio is a web application. Upload a lecture, or open the built-in sample, and it produces a timed transcript, structured notes, a practice quiz and translations, each stage streaming its progress to the browser. It runs on an ordinary laptop with no GPU.

**40**
test lectures in four spoken languages

**0.258**
average word error rate

**10**
translation languages

**122**
automated tests passing

Beyond the brief, the application adds these:

- Every output links back to the audio. A note section, a quiz answer or a transcript word is one click from the moment it was spoken.
- Search works across scripts by sound, so the English word "gradient" finds the same word in Hindi or Tamil text.
- A concept map places the ideas in a lecture along its timeline.
- Theatre mode plays the lecture with word-by-word captions and a rangoli pattern drawn from the loudness of the speech.
- The Benchmark page scores any set of predictions, shows a word-by-word comparison and exports failures as CSV.
- Results are stored by content, so a lecture is only processed once, and a badge on each result says what produced it.

# 4. How it works

**Audio or video** Upload, or the sample › **Silero-VAD** Finds each stretch of speech › **Whisper** Speech to text › **Param-1** Notes and quiz › **NLLB-200** Ten Indian languages

Each stage tries three things in order: a stored result for this exact input, the real model, then an offline method. The studio is a Python backend on Starlette that streams events to a browser frontend written in plain JavaScript modules.

| Stage | Model | Job | Time on the sample |
|---|---|---|---|
| Speech detection | Silero-VAD v5 | Cuts the audio into 26 stretches of speech with timestamps | 7 s |
| Transcription | whisper-hindi-small | Hindi speech to text, one stretch at a time | 2.0 min |
| Notes | Param-1 2.9B Instruct | Structured English notes | 9.6 min |
| Quiz | Param-1 2.9B Instruct | Five multiple-choice questions | 8.2 min |
| Translation | NLLB-200 | Notes and quiz into one language, keeping the answer key | 64 s |

Measured on the 98-second sample, on an Intel i5-1235U with 16 GB of RAM and no GPU. All stages together take about 31 minutes.

### Model selection

The brief says to confirm checkpoint availability, licensing, language coverage, GPU requirements and suitability before using a model. Those checks changed two choices. IndicConformer and IndicTrans2 are gated on Hugging Face and need a personal login, and IndicConformer also needs the NeMo toolkit. The open alternatives run on a CPU with no login: Whisper, with models fine-tuned for Hindi, Tamil and Kannada, and NLLB-200, which uses the same language codes as IndicTrans2. Param-1 is used as the brief asks. The code tries the AI4Bharat models first when their toolkits are installed.

### Search across scripts

Search needs no model. Hindi, Tamil and the other major Indic scripts share one Unicode layout, so a single table reduces them, and Latin spelling, to the same consonant skeleton:

```
gradient  →  K R T N T      ग्रेडिएंट  →  K R T N T      கிரேடியன்ட்  →  K R T N T
```

## 5. The application in use

Every screenshot comes from the running application and shows real model output.

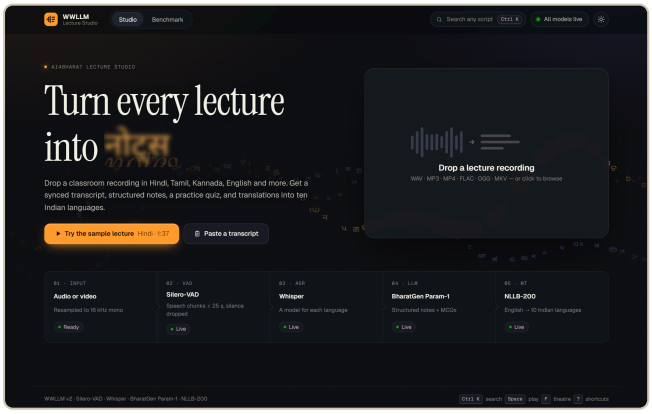



**Figure 1.** The home page. Upload a recording, paste a transcript, or open the sample lecture.

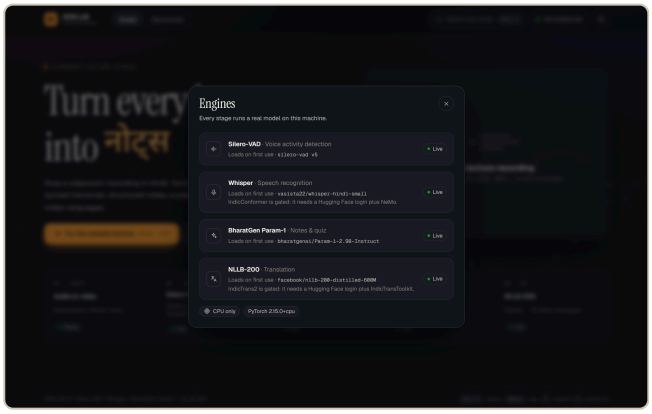



**Figure 2.** The engines panel shows the model behind each stage and whether it can run on this machine.

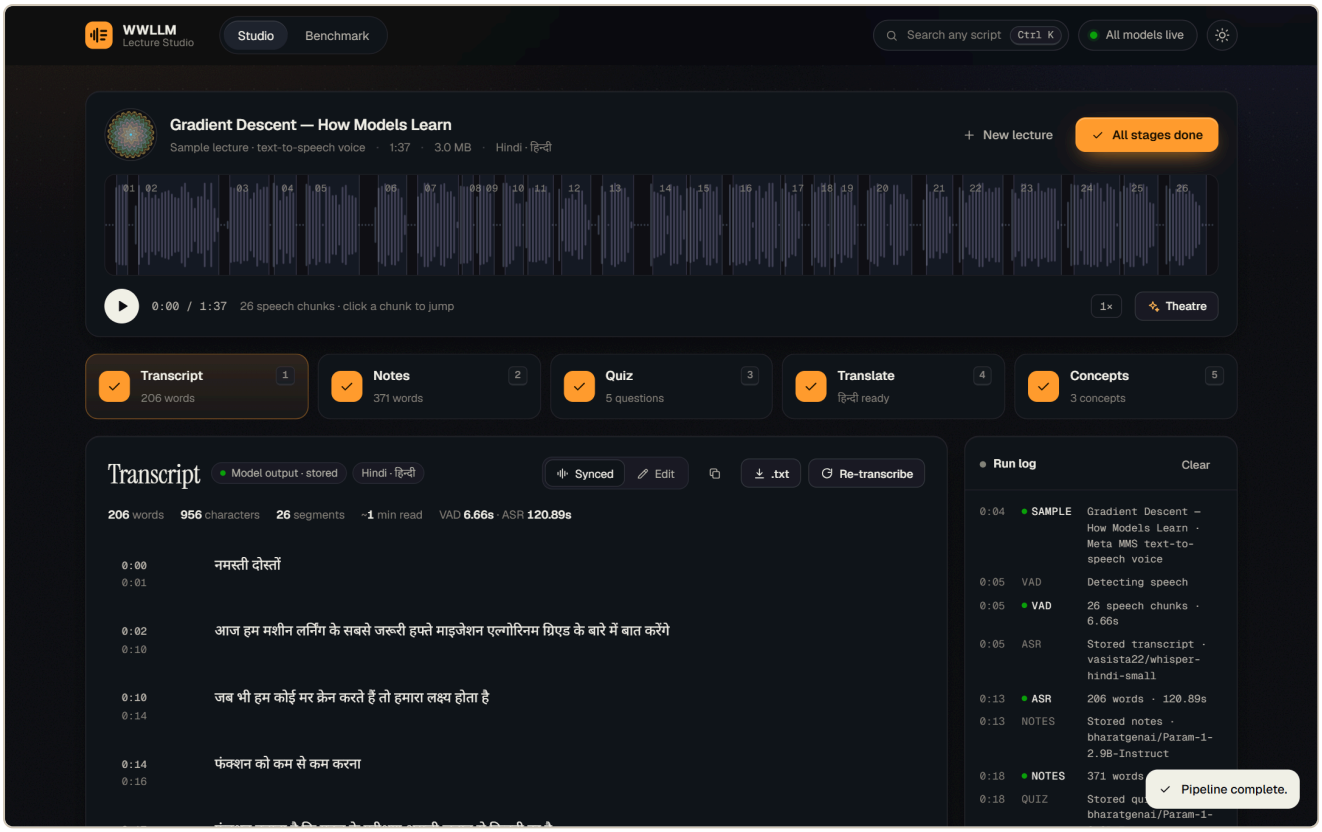



**Figure 3.** The transcript from Whisper, 206 words of Hindi across 26 stretches of speech. Words light up during playback, and clicking one jumps to that moment.




**Figure 4.** Notes from Param-1, which read the Hindi transcript and wrote 371 words of English. Each section has a chip that plays the matching part of the lecture.

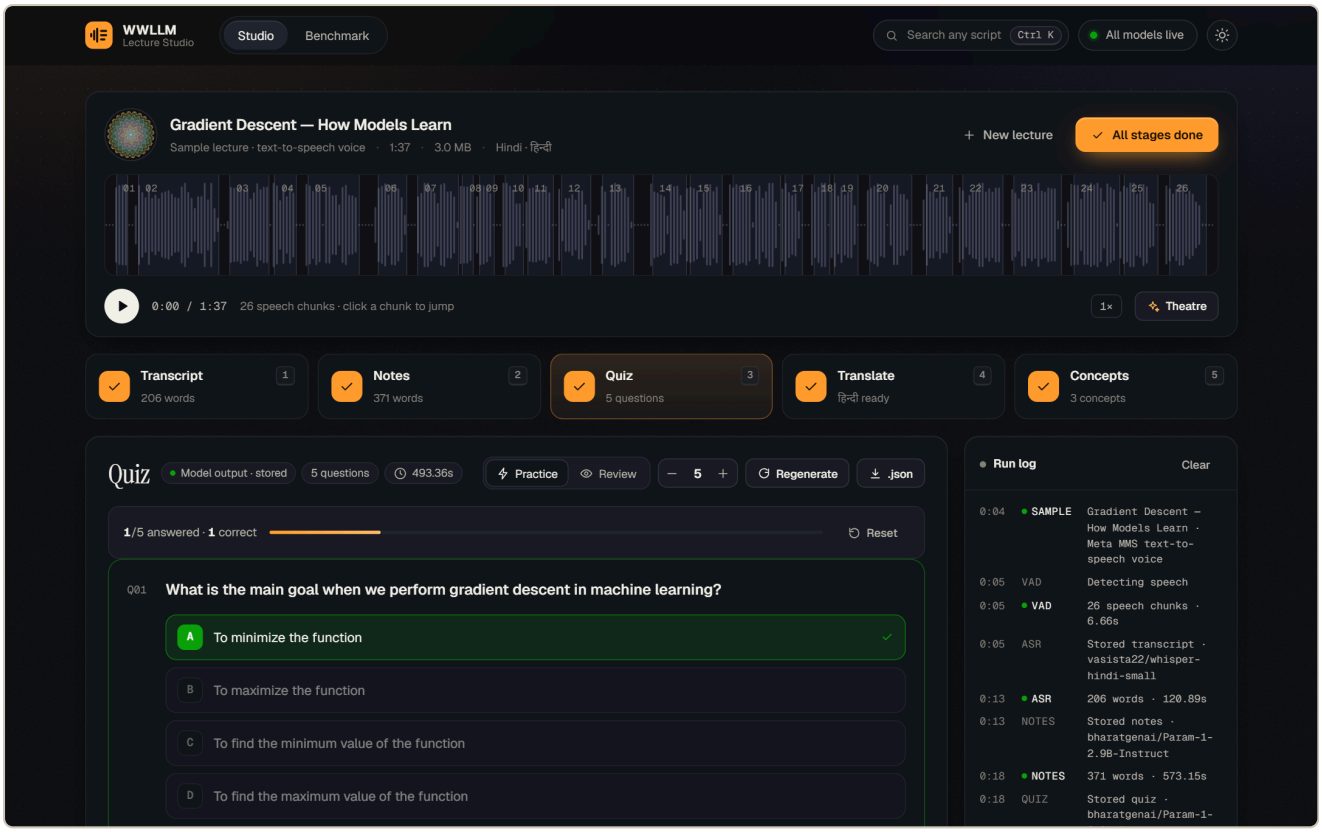



**Figure 5.** The quiz, written by Param-1. After an answer, the student can replay the moment in the lecture that holds it.

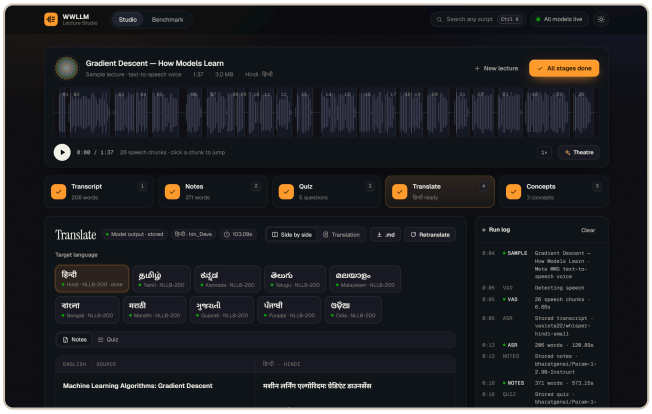


**Figure 6.** Notes and quiz translated into Hindi by NLLB-200. Headings, lists and the answer key are kept.

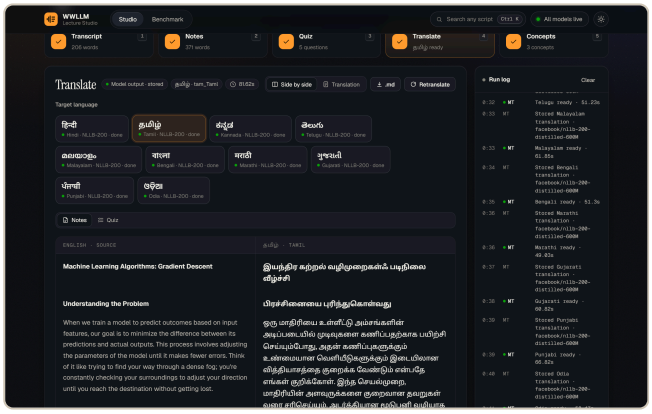


**Figure 7.** The same notes in Tamil, side by side with the English.

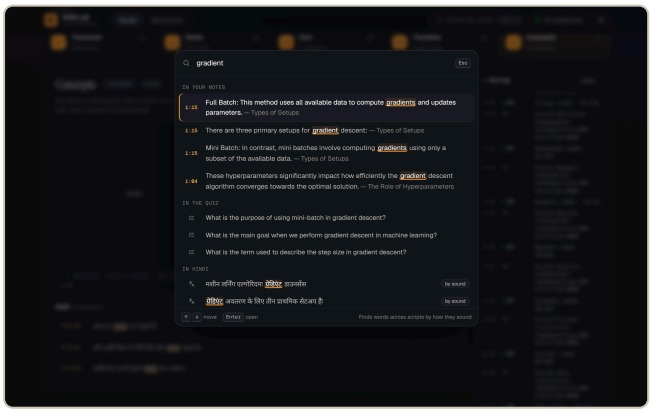



**Figure 8.** Ctrl K search. The English word "gradient" finds matches in the notes, the quiz and the Hindi translation.

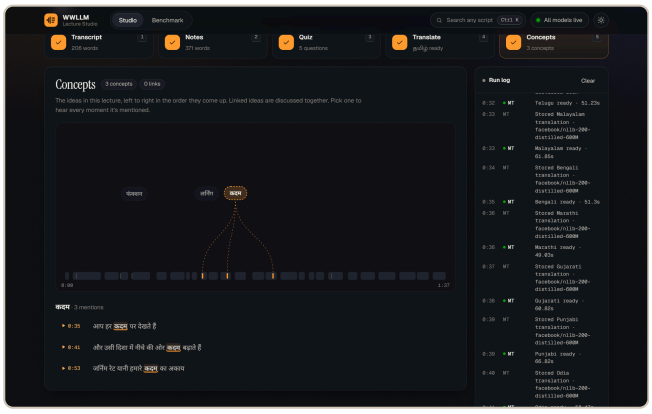


**Figure 9.** The concept map places each idea on the timeline and lists every moment it is mentioned.

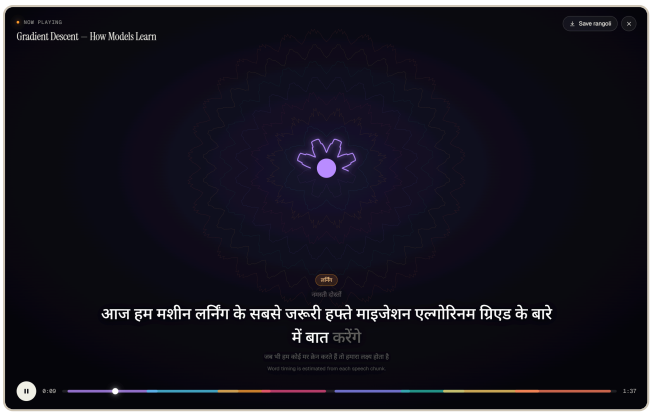



**Figure 10.** Theatre mode, with captions and a rangoli shaped by the loudness of the speech.

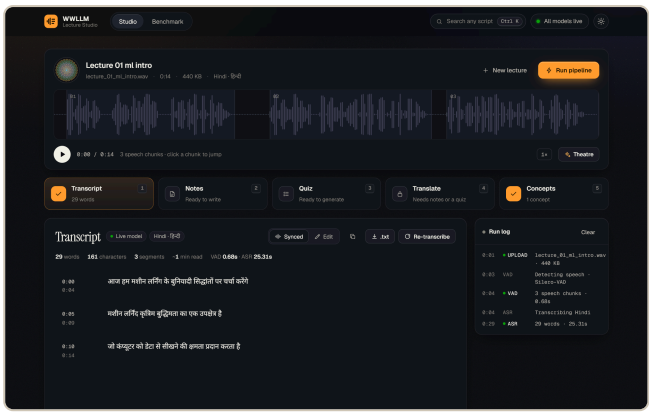



**Figure 11.** A new recording uploaded and transcribed live by Silero-VAD and Whisper.

# 6. Evaluation: word error rate

The test set is 40 short technical lectures. Meta's MMS text-to-speech voices read each reference text aloud, the pipeline transcribed the audio, and the output was scored against the text. Word error rate (WER) counts the words changed, dropped or added, as a share of the reference. Punctuation and case are ignored.

| Language | Model | Lectures | Average WER | Failures |
|---|---|---|---|---|
| English | openai/whisper-small | 22 | 0.257 | 6 |
| Hindi | vasista22/whisper-hindi-small | 14 | 0.277 | 5 |
| Tamil | vasista22/whisper-tamil-small | 2 | 0.225 | 0 |
| Kannada | vasista22/whisper-kannada-small | 2 | 0.167 | 1 |
| **All** | | **40** | **0.258** | **12** |

A lecture fails above a WER of 0.30, so 28 pass and 12 fail. Word accuracy is 77.2%: 619 of 802 reference words. This is clean, machine-read speech, so a real classroom with echo and noise is likely to score worse. The Tamil and Kannada rows rest on two clips each, which is too few to compare languages.




**Figure 12.** The Benchmark page, built from this measurement.

# 7. Failure analysis

Of the 207 word errors, 150 are substitutions (72%), 33 are deletions (16%) and 24 are insertions (12%). The worst lecture is Cache Coherence at 0.684, where "cache coherence protocols like MESI ensure" came out as "KakiCarns protocols. Light mess I endure". Going through the 12 failures shows three causes:

| Cause | How much it matters | Example |
|---|---|---|
| Technical terms and names misheard | Every failing lecture has at least one | "SHA-256" heard as "Cheveth", and "कुबेरनेट्स" as "केनरी" |
| Opening words lost or garbled | The first word goes wrong in 24 of 40 clips | "बैकप्रोपेगेशन एल्गोरिथम" heard as "प्रोपेशन गोरी" |
| Compound words spelled differently | Counted as errors though the meaning is right | "pre-image" as "preimage" is 4 of lecture 28's 7 errors |

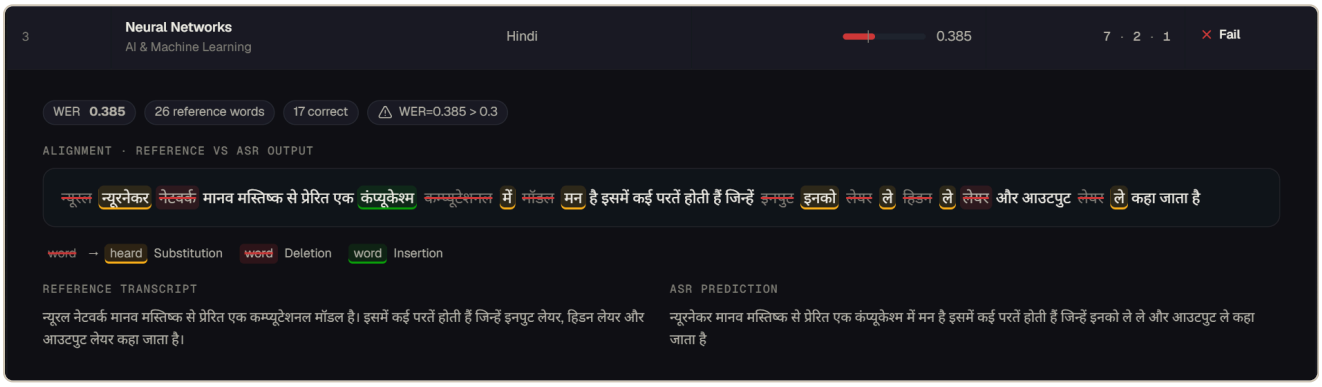



**Figure 13.** Any lecture opens into a word-by-word comparison of the reference and what the model heard.

One fix was tried for the opening words: half a second of silence added before each chunk, on all 14 Hindi clips. Some opening words came back, but the average WER stayed at 0.277 and one more clip failed, so the change was left out. Two fixes remain untested: a custom vocabulary for technical terms, and scoring compound words as one word.

# 8. Evaluation: notes and quiz

The notes and quiz for the sample lecture were checked by hand against its reference text. This covers one lecture, so the figures show how the system behaves and aren't a measured rate.

## Factual consistency

| Statements in the notes | Count | Example |
|---|---|---|
| Supported by the lecture | 8 | A high learning rate overshoots, and a low one is slow |
| Correct, but added from general knowledge | 8 | Batch size as a hyperparameter, which the lecture never mentions |
| Wrong | 1 | Mini-batches are "less efficient than full batches" |

So 16 of 17 statements are correct, working from a transcript with a WER of 0.27. The notes name the topic correctly and keep the lecture's fog analogy. They miss that mini-batch is the most common choice in deep learning, and they drop the lecturer's definition of the gradient as the slope. Notes need a human read before they are trusted.

### MCQ quality

| Question | On topic | Answer key | Explanation |
|---|---|---|---|
| 1. Main goal of gradient descent | Yes | Correct, but a second option says the same thing | None |
| 2. Term for the step size | Yes | Correct: learning rate | None |
| 3. Which is not a type of dataset | Yes | Wrong: confuses datasets with methods | None |
| 4. Where mini-batch is commonly used | Yes | Circular: the question gives away the answer | Present |
| 5. Purpose of mini-batch | Yes | Not supported by the lecture | Contradicts itself |

All five questions are on topic, but only question 2 is fully sound, and a teacher should review the quiz before using it. The likely cause is that Param-1 is writing from a noisy Hindi transcript. Generating the quiz from the English notes is the next thing to try.

### Translation

Translation is optional in the brief and has no formal score here. NLLB-200 gives readable text but garbles technical terms: "Gradient Descent" became ग्रेडिएंट डाउनसेंस in Hindi, close in sound but not the real term.

## 9. Testing

122 automated tests pass: 87 in Python, covering the pipelines, service layer, HTTP API and Streamlit app, and 35 in JavaScript, covering search, anchors and concepts. Without the ML stack installed, the Python suite still passes, with 16 model tests skipped. A fresh clone of the GitHub branch passes the same tests.

A scripted run in Chrome went through the sample lecture, all ten translations, search, Theatre, the Benchmark page and a live upload. All 21 checks passed, with no console or network errors.

# 10. Problems found and fixed

Running real models on real speech exposed these faults.

| Symptom | Cause | Fix |
|---|---|---|
| Models used one CPU core | Importing silero_vad pins the whole process to one thread | Restore the thread count after the import |
| Param-1 crashed on load, then wrote gibberish | Its custom code predates transformers 5, which left the rotary position tables uninitialised | Reset the config and rebuild the tables after loading |
| Each new word re-read the whole prompt | generate() can't use the model's cache format | A small sampling loop that carries the cache itself |
| Whisper kept 80 words of a 206-word lecture | The fine-tuned model stops at the first pause, and chunks held several sentences | One chunk per stretch of speech, which took the sample's WER from 0.77 to 0.27 |
| Notes had no contents list or section chips | Param-1 wrote #### headings, and the interface looked for ## | Level the heading depth on display |
| The quiz failed three ways | Param-1 wrote broken JSON, copied the example, then stopped after one question | A plain text format, with each question started for the model |
| The benchmark was simulated | Hand-written predictions scored against sine-wave audio | Real speech for all 40 clips and a real measurement |

# 11. Repository and running it

Repository: github.com/Tamishsri/WWLLM-CIE-3
Full project: github.com/Tamishsri/WWLLM-CIE-3/tree/feat/lecture-studio

```
pip install -r requirements.txt
python server.py --open
```

Open `http://127.0.0.1:8000` and press Try the sample lecture. Models download on first use, about 8 GB in all. With `requirements-web.txt` alone, the studio still runs the sample from stored results. The README explains how to switch to IndicConformer and IndicTrans2 with a Hugging Face login.

# 12. Limits

- IndicConformer and IndicTrans2 have not been run or measured.
- The accuracy figures come from synthetic speech, so they show the pipeline working and don't predict classroom accuracy.
- Factual consistency and MCQ quality were checked by hand on one lecture.
- Param-1 is slow without a GPU, at about 9.6 min for one set of notes. The sample plays instantly from stored results, but a new lecture takes this long.
- The user picks the spoken language. There is no automatic detection, and the wrong choice produces nonsense.
- Word timing is estimated from chunk timing, so caption highlighting is close but not exact.
- MMS voices and NLLB-200 are CC-BY-NC, which rules out commercial use.